

MAT 1013: Discrete Mathematics

Session 3: Predicates and Quantifiers

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Learning Outcomes

By the end of this session, students should be able to:

1. Distinguish predicates from propositions
2. Identify free and bound variables
3. Interpret quantified statements
4. Translate English statements into quantified form
5. Work with multiple quantifiers
6. Correctly negate quantified statements
7. Explain how domains affect truth values

Part 1: Propositions vs. Predicates

For each of the following statements, discuss:

- Which statements depend on variables?
- Which statements already have truth values?
- Which statements require additional information?
- How to classify these statements?

1. $2 + 5 = 7$
2. $x + 3 = 10$
3. Every student likes mathematics.
4. Some birds cannot fly.
5. n is an even number.
6. $x^2 + y^2 \geq 0$

Statements such as $x + 3 = 10$ are not propositions because their truth value depends on the values of the variables.

Part 2: Predicates and Domains

Definition

A **predicate** is a sentence containing one or more variables that becomes a proposition when values are assigned to the variables.

Examples: $P(x) : x + 3 = 10$

$Q(n) : n \text{ is even}$

Domains of Discourse

The set from which variables are chosen is called the **domain of discourse**.

Example:

$$P(x) : x^2 = 1$$

- Over $\mathbb{N}$: true only for $x = 1$
- Over $\mathbb{Z}$: true for $x = 1, -1$

Activity:

1. For $P(x) : x^2 = 2$, determine the values of x for which $P(x)$ is true over the following domains of discourse.

- | | | |
|---------------------------------------|--------------------------|------------------------------------|
| (a) $\mathbb{N} = \{1, 2, 3, \dots\}$ | (d) $\mathbb{R}$ | (g) $\{x \in \mathbb{R} : x > 0\}$ |
| (b) $\mathbb{Z}$ | (e) $\mathbb{C}$ | (h) $\{x \in \mathbb{R} : x < 0\}$ |
| (c) $\mathbb{Q}$ | (f) $\{2, 4, 6, \dots\}$ | (i) $\{x \in \mathbb{R} : x < 0\}$ |

2. **Domain Matters:** For each predicate, determine whether it is true in different domains $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{R}$.

- | | | |
|---------------|------------------|-----------------|
| (a) $x^2 = 4$ | (b) x is prime | (c) $x + 1 = 0$ |
|---------------|------------------|-----------------|

Part 3: Introduction to Quantifier Symbols

Universal Quantifier

Universal Quantifier

The symbol: $\forall$
means : *for all*, *for every*

Example: $\forall x \in \mathbb{R}, x^2 \geq 0$

Read as:

“For every real number x , x^2 is greater than or equal to zero.”

Existential Quantifier

Universal Quantifier

The symbol: $\exists$
means : *there exists*, *for some*

Example: $\exists x \in \mathbb{Z}, x^2 = 4$

Read as:

“There exists an integer x such that $x^2 = 4$.”

Quantifiers convert predicates into propositions.

Examples:

1. $x > 0$ is not a proposition. Convert it to a proposition using quantifiers. Discuss its truth value over suitable domains of discourse.

Note that there are two ways to quantify this predicate. Also, there are so many different domains of discourse one can consider.

2. There are eight different ways of quantifying the predicate $x + y = 2$ over the reals (i.e., $x, y \in \mathbb{R}$). List them all and discuss the truth value of each.
3. Discuss the truth values of the following quantified statements.
 - (a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \quad y > x$
 - (b) $\exists y \in \mathbb{R} \quad \forall x \in \mathbb{R}, \quad y > x$

Part 4: Free and Bound Variables

Definitions

A variable is called **bound** if it is controlled by a quantifier.
A variable is called **free** if it is not controlled by a quantifier.

Examples:

1. In $\forall x (x + y > 0)$
 - x is bound
 - y is free
2. In $\exists y \forall x (x + y > 0)$, Both variables are bound.

Expressions with free variables are generally not propositions.

Part 5: Universal and Existential Quantifiers

Activity:

1. Classify statements as universal or existential.
 - (a) Every student submitted the assignment.
 - (b) Some students submitted the assignment.
 - (c) All birds have wings.
 - (d) There exists a prime number greater than 100.
2. Translate English Statements in to its symbolic form.
 - (a) Every natural number is positive.
 - (b) Some real numbers are irrational.
 - (c) Every real number has an additive inverse.
 - (d) There exists an integer greater than 100.
 - (e) Not every real number is rational.

3. Express the following in English language.

(a) $\forall x \in \mathbb{R}, x^2 \geq 0$

(b) $\exists x \in \mathbb{Z}, x^2 = 16$

(c) $\forall n \in \mathbb{N}, n + 1 > n$

(d) $\exists x \in \mathbb{R}, x^2 = -1$

(e) $\forall x \in \mathbb{R}, (x > 2 \implies x^2 > 4)$

(f) $\exists n \in \mathbb{N}, n \text{ is even}$

(g) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x + y = 0$

(h) $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, x + y = 0$

(i) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y > x$

(j) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x \geq y$

Part 6: Multiple Quantifiers

Motivation

Many mathematical statements involve several variables.

Example:

$$x + y = y + x$$

To state this formally:

$$\forall x \forall y (x + y = y + x)$$

Examples

1. $\forall x \in \mathbb{R} \exists y \in \mathbb{R} (x + y = 0)$

Read as:

For every real number x , there exists a real number y such that $x + y = 0$.

2. $\exists y \in \mathbb{R} \forall x \in \mathbb{R} (x + y = 0)$

Read as:

There exists a real number y such that for every real number x , $x + y = 0$.

Changing the order of quantifiers can completely change the meaning.

Part 7: Dependent and Independent Variables Under Quantification

Motivation

Consider the statement:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y = 0$$

This statement is **true**. Notice that the value of y depends on the chosen value of x .

For example:

- If $x = 2$, then $y = -2$
- If $x = -5$, then $y = 5$

Thus, the existentially quantified variable y depends on the universally quantified variable x . However, the statement:

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, x + y = 0$$

is **false**. It is false because now y is a constant and there is no single y that works for all x .

Definition

In a quantified statement, a variable is called **dependent** if its value depends on previously quantified variables.

A variable is called **independent** if its value can be chosen without depending on other quantified variables.

Examples

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y = x + 1$

Here:

- x is independent
- y depends on x

For each value of x , a suitable value of y must be chosen.

2. $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, y = x + 1$

Here:

- y must be chosen first
- y cannot depend on x

This statement is false because no single real number y satisfies the equation for all x .

3. $\forall y \in \mathbb{R}, \forall x \in \mathbb{R}, y = x + 1$

Here:

- x and y are arbitrary and hence independent.

This statement is false because the identity does not hold for every choice of x and y .

4. $\exists y \in \mathbb{R}, \exists x \in \mathbb{R}, y = x + 1$

Here:

- x and y are fixed constants.

This statement is true because the identity does hold for suitably chosen x and y . For example, $x = 0$ and $y = 1$.

Key Observation

The order of quantifiers determines which variables may depend on others.

In general:

$$\forall x \exists y P(x, y)$$

means:

*“For every x , we may choose a **corresponding** y .”*

Thus, $y = y(x)$, meaning that it may depend on x .

But:

$$\exists y \forall x P(x, y)$$

means:

“There exists a single fixed y that works for all x .”

So y cannot depend on x . y is a constant.

Examples:

1. Discuss the truth value of the following quantified statements.

(a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y > x$

(b) $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, y > x$

2. For each statement:

- Identify dependent and independent variables
- Decide whether the statement is true or false

(a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y = 0$

(b) $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, x + y = 0$

(c) $\forall n \in \mathbb{N}, \exists m \in \mathbb{N}, m > n$

(d) $\exists m \in \mathbb{N}, \forall n \in \mathbb{N}, m > n$

Common Misconceptions

Misconception	Correction
Quantifier order unimportant	Order determines dependency
Existential variables always fixed	They may depend on previous variables
Variables independent by default	Dependencies arise from quantifier order

Part 8: Negation of Quantified Statements

Key Rules

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$$

Examples

Negate:

$$\forall x \in \mathbb{R}, x^2 \geq 0$$

Solution:

$$\exists x \in \mathbb{R}, x^2 < 0$$

Negate:

$$\exists x \in \mathbb{Z}, x > 10$$

Solution:

$$\forall x \in \mathbb{Z}, x \leq 10$$

Negating Multiple Quantifiers

Negate:

$$\forall x \exists y P(x, y)$$

Solution:

$$\exists x \forall y \neg P(x, y)$$

Activity:

Give a concrete example to demonstrate $\neg(\forall x P(x))$ and $\forall x \neg P(x)$ are not logically equivalent.

Reflection and Exit

Common Misconceptions

Misconception	Correction
Predicates are propositions	Need variable assignment
Quantifier order irrelevant	Order changes meaning
Negation keeps same quantifier	Quantifier switches
Free variables harmless	They prevent proposition formation

Reflection Questions

- Why are quantifiers important?
- Why does quantifier order matter?
- What role do free variables play?

Independent Study

1. Translate:
 - (a) Every integer has a square.
 - (b) Some rational numbers are integers.
 - (c) Every real number has an additive inverse.
2. Identify free and bound variables:
 - (a) $\forall x(x + y > 0)$
 - (b) $\exists z(x + z = 1)$
 - (c) $\forall x \forall y(x + y = y + x)$

- (d) $\forall x (x + y > 0)$
- (e) $\forall x \exists y (x + y = 0)$

3. Negate:

- (a) $\forall x P(x)$
- (b) $\exists x Q(x)$
- (c) $\forall x \exists y P(x, y)$

4. Translate English Statements in to its symbolic form.

- (a) Every natural number is positive.
- (b) Some real numbers are irrational.
- (c) Every real number has an additive inverse.
- (d) There exists an integer greater than 100.
- (e) Not every real number is rational.

5. Express the following in English language.

- | | |
|---|---|
| (a) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x \geq y$ | (g) $\forall x \in \mathbb{Z}, (x \text{ is even} \rightarrow x^2 \text{ is even})$ |
| (b) $\forall x \in \mathbb{N}, (x \text{ is prime} \rightarrow x > 1)$ | (h) $\forall x \in \mathbb{R}, (x \neq 0 \rightarrow \exists y \in \mathbb{R}, xy = 1)$ |
| (c) $\exists x \in \mathbb{Q}, x^2 = 2$ | (i) $\exists n \in \mathbb{N}, \forall m \in \mathbb{N}, n \geq m$ |
| (d) $\forall x \in \mathbb{R}, \forall y \in \mathbb{R}, x + y = y + x$ | (j) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y^2 = x$ |
| (e) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, xy = 1$ | (k) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x + y = y$ |
| (f) $\exists x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y = 10$ | |

6. Negate the statements in the above activity.